

Building 204 - AHUs S1 through S6 Sequence of Operation

Modes: (highest priority to lowest)

The unit shall index to one of six modes:

Fire Mode

Freeze Mode

Smoke Control Mode

Occupied Mode

Bypass Mode

Unoccupied Mode

Fire Mode: (if smoke detectors are present and monitored by the controller)

IF:

- SA smoke is detected
- OR RA smoke is detected

... the controller shall index the unit to "Fire Mode":

The SA Fan and RA Fan will stop

The OA, RA, and EA dampers shall close (if present and controllable)

All Circulation Pumps shall de-energize (if present and controllable)

An alarm will be generated.

Freeze Mode: (if low temperature detector is present and monitored by the controller)

IF:

- NOT in "Fire Mode"
- AND the low temperature detector is true

... the controller shall index the unit to "Freeze Mode":

The SA Fan and RA Fan will stop

The OA and EA dampers shall close

The RA damper shall open

The chilled water valve shall fail closed

The pre-heat valve shall fail open (if present and controllable)

All Circulation pumps will run (if present and controllable)

An alarm will be generated.

Smoke Control Mode:

IF:

- NOT in "Fire Mode" or "Freeze Mode"
- AND commanded by the operator

... the controller shall index the unit to "Smoke Control Mode":

The RA Fan will run at full speed

Duct static pressure control will be enabled with a fixed 2 inch W. C. setpoint

The OA and EA dampers will open fully

The RA damper will close

Occupied Mode:

IF:

- NOT in “Fire Mode”, “Freeze Mode”, or “Smoke Control Mode”
- AND the unit’s master schedule is set for occupancy based on time and date

OR IF:

- Not in “Fire Mode”, “Freeze Mode”, or “Smoke Control Mode”
- AND unit control is overridden by the operator

... the controller shall index the unit to “Occupied Mode”:

SEE BELOW

Bypass Mode:

IF:

- NOT in “Fire Mode”, “Freeze Mode”, “Smoke Control Mode”, or “Occupied Mode”
- AND any terminal unit signals that it’s zone temperature is outside of it’s unoccupied setpoint

OR IF:

- Not in “Fire Mode”, “Freeze Mode”, “Smoke Control Mode”, or “Occupied Mode”
- AND any terminal unit signals “Optimal Start” must begin.

... the controller shall index the unit to “Bypass Mode”:

Minimum OA Control is NOT allowed

If the economizer is disabled, the OA and EA dampers will fully close and the RA damper will fully open.

RA Plenum Static Control is inactive.

The RA Fan speed will be equal to the SA Fan speed plus or minus an offset (adj), and scale factor (adj) as set by the balance contractor.

Unoccupied Mode:

IF:

- NOT in “Fire Mode”, “Freeze Mode”, “Smoke Control Mode”, “Occupied Mode”, or “Bypass Mode”

... the controller shall index the unit to “Unoccupied Mode”.

The SA Fan and RA Fan will stop.

The OA and EA damper will close.

The RA damper will open.

All control valves will close.

Startup Sequence of Operation:

The startup sequence will initiate when the Supply Air Fan transitions from off to on (except in "Smoke Mode").

The OA and EA dampers will remain closed. The RA damper will remain open. The SA and RA fans will be started at the same time and be commanded to minimum speed. Once the SA and RA fans speed feedback signals are greater than 10%, the fans will be proven and the SA Duct Static pressure control, the SA temperature control, and the Cooling Coil Control will run.

Note: Economizer control and Min OA control are disabled during startup.

The SA Duct Static pressure control speed signal will then be allowed to increase the SA fan VFD speed as necessary to reach the initial SA Duct Static set point of 0.250 in wc (adj.). This set point will increase during startup if there is one or more cooling requests (see SA Duct Static pressure optimization section). During startup the RA fan speed will track the SA fan speed with a scale (adj.) and offset (adj.) set by the balance contractor. When the SA Duct Static pressure is within 5 % of set point startup will be complete. Then the RA fan speed will be controlled based on whether economizer or min OA mode is active.

SA Fan:

The SA fan shall run anytime the unit is in "Occupied Mode", "Bypass Mode", or "Smoke Control Mode", unless shutdown on safeties.

The SA Fan shall stop anytime the unit is in "Fire Mode", "Freeze Mode", or "Unoccupied Mode".

To prevent short cycling, the fan shall have a minimum runtime of 5 minutes (adj).

The minimum runtime will not apply if the unit enters "Freeze Mode" or "Fire Mode".

The controller will declare SA Fan status to be true when the SA Fan VFD speed feedback signal (if present) is greater than 10%.

Alarms shall be provided as follows:

- SA Fan Failure or locally switched Off: Commanded on, but the status is off.
- SA Fan in Hand: Commanded off, but the status is on.

RA Fan:

This fan shall run whenever the SA fan runs, unless shutdown on safeties.

To prevent short cycling, the fan shall have a minimum runtime of 5 minutes (adj).

The minimum runtime will not apply if the unit enters "Freeze Mode" or "Fire Mode".

If a new VFD has been installed, existing vortex dampers shall be locked open.

The controller will declare RA Fan status to be true when the SA Fan VFD speed feedback signal (if present) is greater than 10%.

Alarms shall be provided as follows:

- RA Fan Failure or locally switched Off: Commanded on, but the status is off.
- RA Fan in Hand: Commanded off, but the status is on.

MA Stir Fan: (if present and controllable)

This fan shall run whenever:

- The SA fan is running
- AND the OA damper is NOT fully closed
- AND the OA temperature is less than 45 degrees F (adj).

To prevent short cycling, the fan shall have a minimum runtime of 5 minutes (adj).

Pre-Heat Circulation Pump: (if present and controllable)

The Pump will run if the unit is in "Freeze Mode" or the OA temperature is below 45 degrees F (adj).

To prevent short cycling, the pump shall have a minimum runtime of 5 minutes (adj).

Chilled Water Circulation Pump: (if present and controllable)

The Pump will run if the unit is in "Freeze Mode" or the OA temperature is below 45 degrees F (adj).

To prevent short cycling, the pump shall have a minimum runtime of 5 minutes (adj).

Re-Heat Circulation Pump: (if present and controllable)

The Pump will run if the unit is in "Freeze Mode" or the OA temperature is below 45 degrees F (adj).

To prevent short cycling, the pump shall have a minimum runtime of 5 minutes (adj).

SA Temperature Control:

IF:

- The SA Fan status is on

... SA temperature control will be enabled.

The SA Temperature setpoint shall be optimized based on cooling requests from terminal units (zones).

The SA temperature setpoint will be raised or lowered automatically between and upper 65 degrees F (adj) and lower 55 degrees F (adj) limit.

Terminal units shall signal a cooling demand of "1" if their cooling PID is greater than 80%.
Terminal units shall signal a cooling demand of "0" if their cooling PID is less than 80%.
All Terminal units (served by this unit) will have their cooling demands summed.

If the sum is one or more, for 1 minutes (adj), the SA temperature setpoint will be lowered by 0.250 degrees F (adj).

If the sum is zero for 1 minutes (adj), the SA temperature setpoint will be raised by 0.250 degrees F (adj).

On unit start, the initial SA temperature setpoint shall be 60°F (adj).

Alarms shall be provided as follows:

- High SA Temp: If the SA temperature is 5°F (adj) greater than setpoint for 10 min (adj).
- Low SA Temp: If the SA temperature is 5°F (adj) less than setpoint for 10 min (adj).

Economizer Control:

IF:

- SA Temperature Control is enabled
- The unit is in "Occupied Mode" or "Bypass Mode"
- AND the SA fan status is on
- AND the outside air enthalpy is less than the return air enthalpy

... economizer control shall be enabled.

The controller shall measure the MA temperature and modulate the OA, RA, and EA dampers in sequence to maintain a setpoint of 2°F (adj) less than the SA temperature setpoint.

Once the economizer control is enabled, economizer control will remain enabled until the outside air enthalpy is greater than the return air enthalpy by a positive hysteresis of 1btu/lb enthalpy (adj) for 15 minutes (adj).

Cool Coil Control:

IF:

- SA Temperature Control is enabled
- AND the unit is in "Occupied Mode", "Bypass Mode", or "Smoke Control Mode"
- AND OA temperature is greater than 47°F (adj)
- AND the economizer is disabled or fully open

Or If:

- The Startup Sequence is running

... cooling coil control will be enabled.

The controller shall measure the SA temperature and modulate the cooling coil valve to maintain the SA temperature setpoint.

SA Duct Static Pressure Control:

IF:

- SA Temperature Control is enabled
- AND the unit is in "Occupied Mode", "Bypass Mode", or "Smoke Control Mode"

... SA duct static pressure control will be enabled.

The controller shall measure duct static pressure and modulate the SA fan VFD speed to maintain an "Effective Static Pressure" setpoint.

The SA fan VFD speed shall not drop below a minimum speed of 16% (adj.).

Static pressure set point shall be optimized identically (but inversely) just as the SA temperature optimization (see above).

The maximum SA Duct Static set point shall be 1.5 inches W. C.

The maximum SA Duct Static set point shall be 0.250 inches W. C.

The startup SA Duct Static Pressure set point shall be 0.250 inches W. C.

Alarms shall be provided as follows:

- High SA Static Pressure: If the SA static pressure is 25% (adj.) greater than setpoint for 10 min (adj).
- Low SA Static Pressure: If the SA static pressure is 25% (adj.) less than setpoint for 10 min (adj).

RA Plenum Static Control:

IF:

- The unit is in "Occupied Mode"

... RA plenum static control will be enabled.

The EA damper (if "Minimum OA Control" is inactive) and then the RA fan VFD speed shall modulate to maintain a return plenum static pressure no greater than 0.05 inch wc (adj) in the ceiling plenum, referenced to outside air.

Alarms shall be provided as follows:

- High RA Static Pressure: If the RA static pressure is 25% (adj) greater than setpoint for 30 minutes (adj).

Minimum OA Control:

IF:

- The unit is in "Occupied Mode"
- AND the economizer is disabled

... minimum OA control will be enabled.

The OA damper shall index to the adjustable minimum position (as determined by the balance contractor) and stay fixed.

The return air damper will modulate to maintain an adjustable static pressure differential setpoint, across the OA damper of 0.25 inch W. C. (as determined by the balance contractor).

The stroke of the return damper actuator in this mode shall be software limited to an adjustable minimum closed position (as determined by the balance contractor) to avoid duct collapse.

If the RA damper is at 100%, and the static pressure differential across the OA damper is above setpoint, the RA Fan speed will modulate up to maintain setpoint.

A high static pressure switch in the exhaust duct located at the exhaust air dampers shall shutdown the return fan if the pressure rises above 0.75 in. w.g. (adj).

Pre-Heat Control: (if present and controllable)

Pre-heat control is always enabled.

The Pre-Heat valve will modulate to maintain a Pre-Heat temperature setpoint of 45 degrees F (adj).

Alarms shall be provided as follows:

- High PRHT Temp: If the Pre-Heat temperature is greater than 90°F (adj).

- Low PRHT Temp: If the Pre-Heat temperature is less than 36°F (adj).

Dehumidification Control:

IF:

- Cool Coil Control is active
- AND the RA dew point is greater than 58 degrees F (adj)

... dehumidification control will be enabled.

The controller shall override the "SA temperature setpoint optimization" to affect a fixed SA temperature setpoint of 55 degrees F.

Dehumidification will be disabled when the RA dew point less than the 58 (adj) setpoint for 15 min (adj).

Humidification Control:

IF:

- The unit is in "Occupied Mode" or "Bypass Mode"
- AND the SA Fan status is on
- AND the humidifier HEX is enabled
- AND the OA temperature is below 45°F (adj)
- AND the chilled water valve is closed

... humidification control will be enabled.

The controller shall measure the RA humidity to modulate the existing pneumatic humidifier control valve to maintain a RA humidity setpoint of 30 % (adj).

MA Temperature:

The controller shall monitor the MA temperature.

Alarms shall be provided as follows:

- High MA Temp: If the MA temperature is greater than 90°F (adj).
- Low MA Temp: If the MA temperature is less than 36°F (adj).

RA Humidity:

The controller shall monitor the RA humidity.

Alarms shall be provided as follows:

- High RA Humidity: If the RA humidity is greater than 70% (adj).
- Low RA Humidity: If the RA humidity is less than 25% (adj) for 30 minutes (adj).

RA Temperature:

The controller shall monitor the RA temperature.

Alarms shall be provided as follows:

- High RA Temp: If the RA temperature is greater than 90°F (adj).
- Low RA Temp: If the RA temperature is less than 45°F (adj).

END OF SEQUENCE